Rate Limits and Regeneration for the Event Stream of a Moving Edge

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***Abstract*—An event camera reports when the log intensity of a pixel crosses a threshold. When an edge moves across the sensor, every pixel on its path sees the same intensity profile, yet the pixels do not emit the same events: each fires on its own lattice of threshold levels, offset by a threshold phase, and every event carries timing jitter. We determine the rate that these differences require in an analytic model: a ramp edge that passes each pixel once under an affine flow, with independent uniform phases and Gaussian jitter. Given the edge and its motion, a lossless code must still describe the phase and the jitter of every pixel, so its rate per event does not fall as the path grows. Under a mean-squared timing tolerance the rate has three regimes. Below the jitter variance, the code describes every event time, and the rate is exact. Above it, the code describes one number per pixel, and the rate is known within 0.255 bit. Beyond a threshold set by the event spacing of the pixel, the rate given the edge and its motion is zero, and a description of the motion needs a number of nats that grows at most logarithmically with the length of the path. Because a practical sender has the events and not the edge, our codec REMAKE sends the events of one window in place of the edge and moves them along one affine map per block and window. Moving the events instead of regenerating them doubles the timing error of the ideal decoder when the event spacings are equal. On recordings from a static camera and from a moving car, REMAKE reduces the bits by a factor of 1.3 to 1.6 relative to the cheapest scheme that sends a subset of the events at equal fidelity, for groups of four windows, and its advantage shrinks as the time bin of the receiver shortens.**

***Index Terms*—Event camera, send-on-delta sampling, level-crossing sampling, rate-distortion theory, generalized sampling, motion compensation, neuromorphic sensing.**

I. Introduction

An event camera has no frames. Each pixel compares its log intensity with a stored reference and emits an event, with a time stamp and a polarity, whenever the difference reaches a contrast threshold [1], [2]. The output is a stream of address-events whose rate follows the scene: it is close to zero for a static scene and rises with the speed and the contrast of what moves. A sensor node that forwards this stream over a rate-limited link must decide how many bits the stream needs.

The stream of a moving scene looks highly redundant. If an object moves across the sensor, the pixels along the path of one of its edges observe the same intensity profile, one after the other. Yet they do not emit the same events. Each pixel compares the intensity with its own reference, so it fires on its own lattice of levels, and the offset of that lattice, the threshold phase of the pixel, differs from pixel to pixel. The same profile therefore produces a different event train at every pixel (Fig. 1). This paper asks what these differences cost: how many bits a code needs to preserve them, and how many it saves when a timing error is allowed.

The setting has precedents in sampling theory. An event pixel is a send-on-delta sampler [3], which for a monotone input coincides with level-crossing sampling [4], [5]. An object that crosses M pixels is sampled by M such samplers, which recalls the generalized sampling expansion [6], by which a bandlimited signal is recovered from the outputs of m linear systems that are each sampled at $1/m$ of the Nyquist rate. In multi-channel time encoding, several integrate-and-fire encoders with shifted integrators share one input, and the shifts are what make the channels complementary [7]–[9]. The threshold phases play that role here.

Existing event coders approach the rate empirically. Lossless coders exploit the spatial and temporal correlation of addresses and time stamps [10]–[12]. Lossy coders discard or aggregate events [13], [14], and the coder of [15] suppresses events that an on-sensor optical flow can predict and lets the receiver extrapolate them. A recent survey notes that reported results are hard to compare [16], and no accepted distortion measure for lossy event coding exists [17]. On the theoretical side, rate-distortion results for point processes concern Poisson and finite point processes [18]–[21]. They do not model events that a moving profile generates through a threshold rule.

This paper models that mechanism and derives what a code can and cannot save. The phase mechanism holds for any scene (Lemma 1). For the rates we specialize to a ramp edge that passes each pixel once under an affine flow, with independent uniform phases and Gaussian jitter, which isolates the mechanism and gives closed forms. Recordings then test the mechanism in a codec. The central result is the rate-distortion function of the events of a pixel under a timing tolerance. The other results lead to it or follow from it.

1) Mechanism. Given the edge and its motion, the events of a pixel are a function of a single number, its threshold phase, up to timing jitter. The phase is a property of the pixel and is independent across pixels. Next to what the pixels of a path share, the edge and its motion, each pixel therefore holds a private quantity. A lossless code must describe it for every pixel, and its rate per event does not decrease with the number M of pixels on the path (Theorem 1).

2) Rate under a timing tolerance. The tolerance decides which of the private quantities a code must describe (Theorem 2). Below the jitter variance it describes all of them, and the rate is known exactly. Above it one number per pixel remains, and the rate is known within 0.255 bit. Beyond a threshold the rate given the edge and its motion is zero: the decoder regenerates the events. The threshold grows with the event spacing of the pixel, which is the contrast threshold divided by the rate of change of its intensity, so slowly crossed pixels need a larger tolerance. Along a path of M pixels, a

description of the motion of $O(\ln M)$ nats then replaces a lossless description of order M.

3) Regeneration in practice. Because a sender has the events and not the edge, our codec REMAKE replaces the edge by the events of a key window, which the receiver moves along fitted affine maps. The sender finds its regions from the events and writes a complete stream, whose bytes are the cost. Moved events keep the phases of the pixels they came from. At equal event spacings this doubles the timing error of the ideal decoder (Proposition 2), but it lowers the voxel fidelity of a receiver only by a term of order $1/n$. On the model we measure each step from the ideal decoder to REMAKE. On recordings from a static camera [22] and from a moving car [23], we compare REMAKE with schemes that send a subset of the events, with the same coder and at the same fidelity, so that the comparison isolates the contribution of regeneration.

Section II states the model. Sections III and IV contain the theory. Section V describes REMAKE, and Section VI the measurements. Proofs are in the appendices. Rates are in nats unless bits are named, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $[x]^+ = \max(x, 0)$.

II. Model

We first state the pixel rule, from which the phase mechanism follows for any scene. We then specialize to a ramp edge in uniform passage, for which the rates have closed forms.

A. Pixel

Pixels sit at the lattice points $i \in \mathbb{Z}^2$ of the image plane. Pixel i observes the log intensity $\ell_i(t) = L(i,t)$ of a continuous field L.

(A1) Send-on-delta. Pixel i holds a reference μ_i with $|\ell_i - \mu_i| < C$ at the start. It emits an event at the first time t at which $|\ell_i(t) - \mu_i| = C$, with polarity $p = \operatorname{sign}(\ell_i(t) - \mu_i)$, and then sets $\mu_i \leftarrow \mu_i + pC$. The contrast threshold $C > 0$ is common to all pixels and known to the decoder. The *phase* of the pixel is the number $a_i \in (0, C]$ with $\mu_i \in a_i + C\mathbb{Z}$ at the start.

B. Scene and Motion

A template $g : \mathbb{R}^2 \to \mathbb{R}$ gives the log intensity in object coordinates. At time t the object point u is at the image position $\phi_t(u) = J_t u + b_t$, with $J_t \in \mathbb{R}^{2\times 2}$ invertible, so

$$L(x,t) = g\big(\phi_t^{-1}(x)\big) \tag{1}$$

and pixel i observes

$$\ell_i(t) = g\big(u_i(t)\big), \qquad u_i(t) = J_t^{-1}(i - b_t). \tag{2}$$

The curve u_i is the trajectory of pixel i in object coordinates. The velocity field $v(x,t) = \dot\phi_t(\phi_t^{-1}(x))$ is affine in x and satisfies the brightness constancy relation

$$\frac{\partial L}{\partial t}(x,t) + \langle \nabla L(x,t), v(x,t)\rangle = 0, \tag{3}$$

which follows from differentiating $L(\phi_t(u), t) = g(u)$ in t. We call the motion an *affine flow* when the velocity field is also constant in time,

$$v(x) = \Lambda x + v_0, \tag{4}$$

with generator $\Lambda \in \mathbb{R}^{2\times 2}$. A translation has $\Lambda = 0$. The trace of Λ is the rate at which the object grows or shrinks in area, its antisymmetric part rotates the object, and its symmetric traceless part shears it. At a given instant, any smooth velocity field has the form (4) to first order in x around a point, so (4) is the local model of a general motion over a short time.

C. Edge

(A2) Ramp edge. The template is a ramp across a line with unit normal ν,

$$g(u) = \varrho\big(\langle \nu, u\rangle - c_0\big), \qquad \varrho(z) = \begin{cases} 0, & z \le 0, \\ \gamma z, & 0 \le z \le w, \\ V, & z \ge w, \end{cases} \tag{5}$$

with slope $\gamma > 0$, width w and height $V = \gamma w = nC$ for an integer $n \ge 1$.

Pixel i sees the edge through its edge coordinate

$$z_i(t) = \langle \nu, u_i(t)\rangle - c_0, \qquad \ell_i(t) = \varrho\big(z_i(t)\big). \tag{6}$$

(A2′) Uniform single passage. The edge passes pixel i once: $z_i < 0$ before a time t_i, $z_i > w$ after the passage, and in between z_i is affine in t with slope $r_i > 0$.

Inside the ramp, $\nabla L = \gamma \nu_t$ with $\nu_t = J_t^{-\top}\nu$ a normal of the edge in the image, and $\dot z_i = -\langle \nu_t, v(i,t)\rangle$. By (3), the intensity of pixel i rises during the passage at the rate

$$\gamma r_i = -\langle \nabla L, v\rangle = \|\nabla L\|\, v_{\perp,i}, \tag{7}$$

with the field evaluated at pixel i. Here $v_{\perp,i} = -\langle \nabla L, v\rangle / \|\nabla L\|$ is the speed at which the edge advances over the pixel along its normal, counted positive when the intensity of the pixel rises, as it does in a passage with $r_i > 0$. We call

$$\tau_i = \frac{C}{\gamma r_i} = \frac{C}{\|\nabla L\|\, v_{\perp,i}} \tag{8}$$

the *event spacing* of pixel i. It is the time that the ramp takes to pass the pixel, divided by n. At a fixed image slope $\|\nabla L\|$ it is inversely proportional to the normal speed of the edge. Assumption (A2′) holds exactly for a translation, where τ_i is the same for all pixels, and for a plane that approaches the camera along the optical axis at constant speed, where J_t^{-1} is affine in t and b_t is fixed. For an affine flow it holds to first order (Remark 1), and τ_i varies from pixel to pixel.

D. Phase, Jitter and Clock

(A3) Phase. Before the edge arrives, pixel i rests at the level 0 with its reference behind the passage, $\mu_i \in (-C, 0]$, so that $a_i = \mu_i + C$. The phases a_i are independent across pixels, uniform on $(0, C]$, and independent of the jitter.

(A4) Jitter and clock. The kth event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$ variables, independent of the phases, with $\sigma > 0$. The stored value is $Y_{i,k} = \delta\lfloor T_{i,k}/\delta \rfloor$ for a clock resolution δ. The index k of each recorded event is known.

The uniform phase does not arise inside the model, in which every pixel rests at the same level. It stands for the offsets

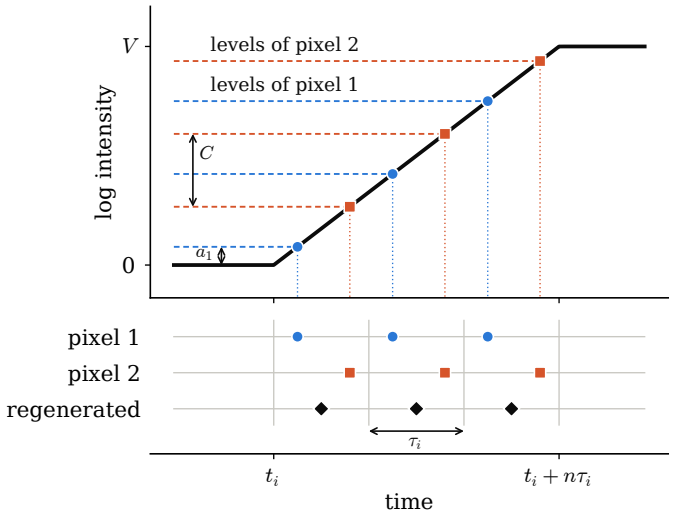


Fig. 1. One ramp, two pixels with different threshold phases, and the regenerated events (22). Time is measured from the arrival of the edge at each pixel. Jitter is omitted.

of real pixels and for their earlier activity. A pixel that the edge crosses from the high level to the low level is covered by the mirrored assumption, with its reference in $[V, V + C)$. Remark 4 treats a reference that is ahead of the passage.

We write $T_i = (T_{i,1}, \dots, T_{i,n})$ and $Y_i = (Y_{i,1}, \dots, Y_{i,n})$, and $\mathcal{P}$ for the set of pixels that the edge passes during the observation, with $M = |\mathcal{P}|$. Let $\eta(a, \sigma)$ denote the differential entropy of the sum of a uniform variable on $(0, a)$ and an independent $\mathcal{N}(0, \sigma^2)$ variable. It satisfies

$$\tfrac{1}{2}\ln\big(a^2 + 2\pi e\sigma^2\big) \le \eta(a,\sigma) \le \tfrac{1}{2}\ln\big(2\pi e(a^2/12 + \sigma^2)\big) \quad (9)$$

by the entropy power inequality [24, Ch. 17] and by the Gaussian bound on the entropy at a given variance [24, Ch. 8], and $\eta(a, \sigma) - \ln a \to 0$ as $\sigma/a \to 0$.

III. Structure and Lossless Limit

Lemma 1. *Under (A1), for any continuous ℓ_i, every event of pixel i occurs at a time t with $\ell_i(t) \in a_i + C\mathbb{Z}$.*

The lattice $a_i + C\mathbb{Z}$ never changes. The phase is a property of the pixel, set by its history and not by the edge that passes.

Lemma 2. *Under (A1), (A2), (A2′) and $\mu_i \in (-C, 0]$ before the passage, pixel i emits exactly n events, all of positive polarity, at*

$$t_{i,k} = t_i + A_i + (k-1)\tau_i, \qquad k = 1, \dots, n, \quad (10)$$

where $A_i = a_i/(\gamma r_i)$. Under (A3), A_i is uniform on $(0, \tau_i]$.

Given the template and the flow, the noise-free events of a pixel are thus a function of one number, its phase. Fig. 1 shows two pixels with $n = 3$ that see the same ramp: the intensity trajectories coincide, and the event trains differ by the difference of the phases. In the object coordinate, an event is a point of the graph of the profile on the level lattice of its pixel: the M pixels of a path sample the same profile on M lattices with independent offsets.

Theorem 1 (Lossless code length)**.** *Under (A1) to (A4) and (A2′), the vectors T_i, $i \in \mathcal{P}$, are independent given the template and the flow,*

$$h(T_i) = \frac{n-1}{2}\ln(2\pi e\sigma^2) + \eta(\sqrt{n}\,\tau_i, \sigma), \quad (11)$$

and

$$H(Y_i \mid g, \phi) = n\ln\frac{1}{\delta} + h(T_i) + e_\delta, \quad (12)$$

where $e_\delta \ge 0$ for every δ, and $e_\delta \to 0$ as $\delta/\sigma \to 0$, uniformly in τ_i.

When $\sigma \ll \tau_i$ and $\delta \ll \sigma$, (12) gives the cost per event

$$\frac{H(Y_i \mid g, \phi)}{n} \approx \frac{n-1}{n}\ln\frac{\sigma\sqrt{2\pi e}}{\delta} + \frac{1}{n}\ln\frac{\sqrt{n}\,\tau_i}{\delta}. \quad (13)$$

One event of each pixel carries the cost of the phase, and the other $n-1$ carry that of the jitter. The code length of the path is the sum of (12) over $\mathcal{P}$. It grows in proportion to M, and neither term of (13) falls as the path gets longer: a decoder that knows the edge and its motion exactly still needs (13) for every event. What the pixels of a path share, the edge and its motion, a code can describe once. What is private to each pixel, its phase and its jitter, it must describe for each pixel, and (12) is that irreducible conditional rate. Section VI-F measures the saving that the motion brings to a code on recordings.

Remark 1 (Accuracy of the uniform passage). For an affine flow (4), $\dot{\nu}_t = -\Lambda^\top \nu_t$ and $\ddot{z}_i = \langle \nu_t, \Lambda v(i)\rangle$, hence $|\ddot{z}_i| \le \kappa_i |\dot{z}_i|$ with $\kappa_i = \|\Lambda\|/|\cos\vartheta_i|$, where ϑ_i is the angle between the velocity $v(i)$ and the normal ν_t. Take r_i as the slope of z_i at t_i. Over a passage of duration $P_i = n\tau_i$, the event times then deviate from (10) by at most $\kappa_i P_i^2/2$, to first order in $\kappa_i P_i$. For a generator of norm $0.3\,\mathrm{s}^{-1}$, a passage of 5 ms and motion along the normal this is $4\,\mu$s, three orders of magnitude below the passage time. The bound grows without limit for edges that are nearly parallel to their motion. Such edges pass few pixels per unit time and contribute few events.

Remark 2 (Several edges). Theorem 1 counts one edge that passes each pixel once. If further edges of the same object pass the same pixel, they meet the same lattice, so the phase is paid once per pixel and not once per edge. With many edges per pixel the cost per event tends to the jitter term $\ln(\sigma\sqrt{2\pi e}/\delta)$, which does not depend on the path either.

IV. Rate Under a Timing Tolerance

The results of this section and the next assume decoders with different inputs. Theorems 2 and 3 count the bits of the event times when the encoder and the decoder know the template, the flow, and hence the pixels that the edge passes and the number of events of each, and when the index of every event is known (A4). Their zero rate is a rate given all of this. Proposition 1 adds the cost of the flow, while the decoder still holds the template. REMAKE (Section V) holds neither: it sends events of a key window in place of the template and maps in place of the flow, and every bit of its stream is counted.

The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is the mean-squared timing error

$$d(T_i, \hat{T}_i) = \frac{1}{n}\|T_i - \hat{T}_i\|^2. \quad (14)$$

A. One Pixel

Let $R_i(D) = \inf I(T_i;\hat{T}_i)$, with the infimum over the conditional laws of $\hat{T}_i$ that satisfy $\mathbb{E}\, d(T_i,\hat{T}_i) \le D$, be the rate-distortion function of the events of pixel i. It is the rate that block codes over many independent passages approach. Define

$$s_i^2 = \frac{n\tau_i^2}{12} + \sigma^2, \qquad D_{A,i} = \sigma^2 + \frac{\tau_i^2}{12}, \tag{15}$$

$$\Gamma_i = \tfrac{1}{2}\ln(2\pi e\, s_i^2) - \eta(\sqrt{n}\,\tau_i, \sigma). \tag{16}$$

By (9), $0 \le \Gamma_i \le \frac{1}{2}\ln(2\pi e/12)$, which is 0.255 bit, for every σ and τ_i.

Theorem 2 (Rate-distortion function of a pixel)**.** *Under (A1) to (A4) and (A2′),*

$$R_i(D) = h(T_i) - \frac{n}{2}\ln(2\pi e D) \qquad \textit{for } 0 < D \le \sigma^2, \tag{17}$$

$$\big[U_i(D) - \Gamma_i\big]^+ \le R_i(D) \le U_i(D) \qquad \textit{for } \sigma^2 < D < D_{A,i}, \tag{18}$$

$$R_i(D) = 0 \qquad \textit{for } D \ge D_{A,i}, \tag{19}$$

with

$$U_i(D) = \frac{1}{2}\ln\frac{s_i^2}{nD - (n-1)\sigma^2}. \tag{20}$$

Moreover,

$$R_i(D) \ge \frac{n\,(D_{A,i} - D)}{2s_i^2} \qquad \textit{for } 0 < D \le D_{A,i}, \tag{21}$$

so that $R_i(D) > 0$ for $D < D_{A,i}$.

The three regimes have a direct reading. Below the jitter variance, every bit of timing precision costs a bit, per event: the rate falls by $\log_2 10 = 3.32$ bits per event and per decade of the root-mean-square tolerance $\sqrt{D}$. Between σ^2 and $D_{A,i}$, the $n-1$ jitter components orthogonal to a common shift of the events are no longer described, and the code describes one number per pixel, the phase together with the mean jitter, to within 0.255 bit. From $D_{A,i}$ on, nothing is described. The decoder that attains (19) outputs the mean times

$$\hat{T}_{i,k} = t_i + \big(k - \tfrac{1}{2}\big)\tau_i, \qquad k = 1,\dots,n. \tag{22}$$

It *regenerates* the events of the pixel from the edge and the flow. Each regenerated event is within $\tau_i/2$ of its noise-free time, and the count and the polarities are exact.

Two of these statements extend beyond the ramp. If the noise-free event times of a pixel are any function of its phase, (17) holds below the jitter variance with the entropy of the actual event times, and the smallest distortion at zero rate is $\frac{1}{n}\operatorname{tr}\operatorname{Cov}(T_i)$, attained by the mean times. The middle regime, in which one number per pixel is described to within 0.255 bit, and the closed forms (11), (20) and (15) belong to the ramp in uniform passage, where the phase shifts all events of a pixel by a common time.

The threshold $D_{A,i}$ grows with the square of the event spacing. By (8), a pixel whose intensity changes slowly has a large spacing and needs a large tolerance before its events can be regenerated, and a pixel that is crossed quickly needs a small one.

B. A Path

For the M pixels of a path, let the distortion be the average of (14) over $i \in \mathcal{P}$, and let $R_{\mathcal{P}}(D)$ be the rate-distortion function of $(T_i)_{i\in\mathcal{P}}$ under it.

Theorem 3 (Rate-distortion function of a path)**.** *Under (A1) to (A4) and (A2′),*

$$R_{\mathcal{P}}(D) = \min\Big\{\sum_{i\in\mathcal{P}} R_i(D_i) :\ \frac{1}{M}\sum_{i\in\mathcal{P}} D_i \le D\Big\}. \tag{23}$$

In particular,

$$R_{\mathcal{P}}(D) = \sum_{i\in\mathcal{P}} h(T_i) - \frac{Mn}{2}\ln(2\pi e D) \qquad \textit{for } 0 < D \le \sigma^2, \tag{24}$$

and $R_{\mathcal{P}}(D) = 0$ if and only if

$$D \ \ge\ \bar{D}_A = \sigma^2 + \frac{1}{12M}\sum_{i\in\mathcal{P}} \tau_i^2. \tag{25}$$

If $\tau_i \le \tau_{\max}$ on the path, then for $D \le \bar{D}_A$

$$R_{\mathcal{P}}(D) \ge \frac{nM\,(\bar{D}_A - D)}{2\,(n\tau_{\max}^2/12 + \sigma^2)}. \tag{26}$$

If the tolerance is imposed on every pixel separately, the rate is zero if and only if $D \ge \max_i D_{A,i}$, and the slowest crossing sets the tolerance. By (26), below $\bar{D}_A$ a rate that is proportional to M remains.

C. The Motion

Theorems 2 and 3 assume that the decoder knows the flow. It remains to count what describing the flow costs. An error in a speed displaces the regenerated events in proportion to the distance that the edge has traveled, so a longer path needs the motion to a finer precision for the same distortion. A parameter of fixed dimension costs only logarithmically more at a finer precision, and Proposition 1 states this in general form.

Proposition 1 (Description of the motion)**.** *Let the flow depend on a parameter $\theta = (\theta_1,\dots,\theta_q)$ that ranges over a box whose jth side has length at most $L_j M^{\alpha_j}$, and let the mean times $\bar{t}_{i,k}(\theta) = t_i(\theta) + (k - \frac{1}{2})\tau_i(\theta)$ of (22) satisfy $|\partial \bar{t}_{i,k}/\partial\theta_j| \le S_j M^{\beta_j}$ on the box, for all events of the path, with constants $L_j, S_j > 0$ and exponents $\alpha_j, \beta_j \ge 0$ that do not depend on M. Then for every $\xi > 0$ the events of the path can be described with average distortion at most $(1+\xi)\bar{D}_A$ using*

$$B \le \Big(\sum_{j=1}^{q}(\alpha_j + \beta_j)\Big)\ln M + O(1) \quad \textit{nats}, \tag{27}$$

where the $O(1)$ term depends on q, ξ, σ and the constants and not on M.

For a translation along a line with a speed in a fixed interval that excludes zero, θ holds the position of the edge, with a range of order M and a sensitivity of order one, and the speed, with a range of order one and a sensitivity of order M, so $B \le 2\ln M + O(1)$. The same holds with finite exponents, and hence $B = O(\ln M)$, for an affine flow whose normal speed is bounded away from zero on the path and whose duration

is polynomial in M. The lossless cost of the same events is the sum of (12), of order M, so the ratio of the two grows at least as $M/\ln M$. Above $\bar{D}_A$ the description that remains is $O(\ln M)$. Below it, a rate proportional to M remains. The bound (27) is for a receiver that holds the edge. Section V accounts for the template and the regions.

Remark 3 (Event count under a change of scale). An edge segment emits n events for every pixel that it passes, so its event rate is n times the area that it sweeps per unit time, that is n times its length times its mean normal speed. Under a similarity flow about a fixed point, growth by a factor ζ makes the segment ζ times longer and ζ times faster, and the rate grows as ζ^2. For an object that approaches at constant speed, the image speed itself grows as ζ^2 and the rate as ζ^3. A regeneration that moves a fixed set of events keeps their number. Under a change of scale it must also change the number of events, or regenerate them from the edge.

Remark 4 (Departures from the model). *Reference ahead of the passage.* If $\mu_i \in (0, C)$ before a rising passage, as after a passage of the opposite polarity at the same pixel, the first level that the intensity meets is the reference itself, which does not fire. The pixel then emits $n-1$ events, and the results hold with $n-1$ in place of n and $t_i + \tau_i$ in place of t_i. *Height.* If the height of the edge is not a multiple of the threshold, $V/C = n+f$ with $0 < f < 1$, pixel i emits $n+1$ events when $a_i \le fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, and a zero-rate statement then holds only for a distortion measure that tolerates one unmatched event per pixel, with a modified threshold. *Sensor.* Noise events, threshold mismatch between pixels, a refractory time and a jitter that depends on the slope are outside the model, and their effect on the three regimes is not analyzed here.

V. A Regenerating Codec

The decoder (22) needs the edge and the flow. A practical sender has neither. It has the events. The codec of this section, REMAKE (REgeneration by Mapping Affine Key Events), replaces the edge by events that the receiver already holds and the flow by maps that the sender fits, and it finds its regions without any annotation.

A. Regions, Key Events and Groups

Time is cut into windows of length T_0, and consecutive windows form groups of K. The sensor is cut into square blocks of $B \times B$ pixels. In the first window of a group, the key window, the blocks that hold at least ψB^2 events are the regions of the group, where the threshold ψ is a density in events per pixel. The sender transmits the events of the regions, the key events, and for each of the following $K-1$ windows one affine map per region. The receiver moves every key event (x, p, t) of a region to the pixel nearest to the image of x under the map of that region and to the time $t+mT_0$, where m counts the windows since the key window, and it keeps the polarity. A moved event may leave its block. One that leaves the sensor is dropped. The moved events are the reconstruction of window m. The regions stay fixed within a group, so what enters the scene after the key window is not reconstructed before the next one.

Under an affine flow, the map from one instant to an instant mT_0 later is the same for all instants of a window, so one map per window is exact for the positions of the edges of a region that moves with the flow. It does not change the number of events, which a change of scale does (Remark 3).

B. Maps and Their Fit

With c the center of the block, a map moves the position x to

$$x' = x + d + \frac{2}{B} A\,(x - c), \tag{28}$$

where $d \in \mathbb{Z}^2$ is a displacement in pixels and $A \in \mathbb{Z}^{2\times 2}$ holds four shape parameters. Each of them is a displacement in pixels at the edge of the block: the diagonal of A stretches the block along the two axes, and the other two entries shear or rotate it. With $A = 0$ the map is a translation.

The sender fits the maps of a window region by region, by a greedy search for the number of moved events that can be paired with true events of that window inside a voxel of one fixed working grid, which is the numerator of the fidelity (30). It visits the regions in the order of decreasing event count, and a true event that an earlier region has paired is not available to a later one, so the pairs of the regions add up to the pairs of the whole reconstruction. For each region a search from coarse to fine finds the displacement, from zero in the first window and from the extrapolated map of the previous window afterwards, and a coordinate search then finds the shape parameters. On the true events that are still unpaired when a region is visited, its map never pairs fewer events than the translation it started from, and that translation never fewer than no motion. Results on other grids use the same maps.

C. Bitstream and Event Coders

A stream is a header with the configuration (the sensor size, B, K, T_0, the kind of map and the event coder), followed by one record per group: the index of its key window, the key events and the maps, each of the two preceded by its length. The regions are not listed. They are the blocks that hold key events, and the maps follow in the raster order of these blocks, as differences along m in a variable-length integer code, compressed with Zstandard [25]. The bits of a group are eight times the bytes of its record. The header, 28 bytes per stream, is less than 10^{-3} of a stream and is left out. A record is complete when the last window of its group has arrived, so the sender delays a stream by one group. Every fidelity of REMAKE in Section VI is measured on events that a decoder rebuilds from the bytes alone.

Two event coders serve the key events and all schemes that we compare with. The *transport coder* keeps time stamps of one microsecond: it packs the addresses, writes the time differences in a variable-length code and applies Zstandard at its first level. The *matched coder* is for a receiver that counts events on a known grid: it sorts the events of a window by

voxel and writes the gaps between voxel indices in a variable-length code, followed by Zstandard at level 19. A subset of the events is then coded on the grid of the receiver. The key events are coded at one pixel and the time bin of that grid, because a map moves them by whole pixels and keeps their time bin.

D. Moved Events Against Regenerated Events

REMAKE differs from the decoder (22) in three ways. It moves the phases of the key pixels where (22) outputs the mean over the phase, it moves the jitter of the key events, and it moves a fixed number of events. The model gives the cost of the first two exactly.

Proposition 2 (Moved events)**.** *Let the same edge pass pixel j in the key window and a different pixel i later, under (A1) to (A4) and (A2′), and let the map be exact, so that the kth event of pixel j is moved to pixel i at the time $\tilde{T}_{i,k} = T_{j,k} + t_i - t_j$. Then*

$$\frac{1}{n}\,\mathbb{E}\|T_i - \tilde{T}_i\|^2 = D_{A,i} + D_{A,j} + \frac{4n^2-1}{12}\,(\tau_i - \tau_j)^2. \quad (29)$$

With equal event spacings the error is $2D_A$, twice that of (22), which is the timing error that REMAKE accepts for not needing the edge. Unequal spacings, as under a change of scale, add the last term, which grows with the square of the number of events per pixel. If the moved events fall on their own pixel, as for an object that is nearly at rest, the phases coincide and only the jitter term $2\sigma^2$ remains. The third difference, the number of events, is not a timing error. Section VI-C measures these differences on the model, together with those that REMAKE adds: maps in whole pixels, windows that cut event trains, and maps that are fitted.

E. Receiver Representation, Fidelity and Baselines

The theory uses the timing error (14), which presupposes that regenerated and true events are paired. For measured streams we use the representation that event-based receivers compute, a voxel grid: the number of events of each polarity in blocks of $\Delta_x \times \Delta_x$ pixels and in time bins of length Δ_t. For a reconstruction with voxel counts c' and a true set with counts c, of sizes N' and N, the fidelity is

$$F = \frac{2\sum \min(c', c)}{N' + N}, \quad (30)$$

the share of the events of the two sets that can be paired one to one inside a voxel. It is computed over the complete sensor, so the events outside the regions count as lost. A subset that holds a share χ of the events has $F = 2\chi/(1+\chi)$ on every grid, whichever events it keeps. A reconstruction of fidelity F is therefore matched by a subset of the share

$$\chi_F = \frac{F}{2 - F}. \quad (31)$$

Schemes that send a subset of the events and regenerate nothing are the baseline. We measure three: a random subset, complete windows with the other windows dropped, and, in every window, the blocks that the rule of the key window selects. Two such schemes can be mixed over windows, which mixes their shares and their bits linearly, so the cheapest subset scheme at a share χ costs $b_{\mathrm{sub}}(\chi)$ bits per event, the lower convex envelope of the measured operating points and the origin. For a codec that uses b bits per event of a group and reaches the fidelity F over the group, the key window included, we report

$$G = \frac{b_{\mathrm{sub}}(\chi_F)}{b}, \quad (32)$$

the factor by which it lowers the bits at equal fidelity. A random subset alone is a weaker baseline than the envelope: its events lie further apart than those of whole windows or whole blocks, and each of them costs more bits.

The fidelity (30) and the timing error (14) are different distortion measures, and the bin length plays the role of the tolerance. One case connects them. For blocks of one pixel, a train that is long against the event spacing, and a jitter that is small against the bin, Lemma 2 gives the fidelity of the regenerated events (22) directly: a bin holds at most one event of each train when $\Delta_t \le \tau_i$, the offset between the two trains is uniform, and in expectation over the phase

$$F = \frac{\Delta_t}{\tau_i} \qquad \text{for } \Delta_t \le \tau_i, \quad (33)$$

with $F = 1$ at $\Delta_t = \tau_i$. The voxel counts of a pixel are thus reproduced only when the bin is at least as long as its event spacing, the same scaling as the threshold of Theorem 2, for which $\sqrt{D_{A,i}} = \tau_i/\sqrt{12}$ when $\sigma \ll \tau_i$. In larger blocks the independent phases of the pixels average, and the fidelity is higher at a given bin length.

On this grid, moved events reach almost the fidelity of regenerated events. With equal spacings and no jitter, the kth moved event lies at the offset $\delta = A_i - A_j$ from the kth true event, and δ has the triangular density $(\tau_i - |\delta|)/\tau_i^2$ on $(-\tau_i, \tau_i)$. Two events at the offset x share a bin with probability $[1 - |x|/\Delta_t]^+$ over the position of the bins. When $|\delta| > \tau_i - \Delta_t$, a moved event can share its bin with a neighbor of its true event instead, which has a partner for $n-1$ of the n events of a train. For a train of n events and $\Delta_t \le \tau_i/2$, the two contributions give

$$F = \frac{\Delta_t}{\tau_i} - \frac{\Delta_t^2}{3n\tau_i^2}, \quad (34)$$

against Δ_t/τ_i for the regenerated events with the same n and Δ_t. A voxel grid does not distinguish the events of a pixel, and the doubled timing error of (29) therefore lowers the fidelity only by a term of order $1/n$.

VI. Experiments

The experiments address three questions: on the model, the three regimes and the cost of each step from the ideal decoder to REMAKE (Sections VI-B and VI-C); on recorded streams, the bits that regeneration saves at equal fidelity (Section VI-D); and the conditions under which this saving shrinks (Section VI-E).

A. Protocol

Data. The recordings from a static camera are the 112 recordings of the training split of eTraM [22], 61 by day and 51 by night, from a sensor of 1280×720 pixels above a road, with 2.4×10^{10} events. The recordings from a moving camera are the 41 training sequences of DSEC [23], left event camera, 640×480 pixels, on a car, with 3.7×10^{10} events in 2634 s. A window of the median DSEC sequence holds 1.15 events per pixel, against 0.04 for eTraM. Each recording is replayed as a stream, in the order of its events.

Codec. Windows have $T_0 = 33.3$ ms. Groups of $K = 2$, 4, 8 and 11 windows share their key window, which is taken every 3 s of an eTraM recording and every 2 s of a DSEC sequence. The working grid has $\Delta_x = 2$ pixels and $\Delta_t = T_0/3 = 11.1$ ms, and the event coder is the matched coder on that grid unless the transport coder is named. Each dataset is calibrated on a tuning subset of its own: twelve eTraM recordings, every tenth by day and by night, and five DSEC sequences, every tenth of the sorted list. On it the block side is selected among 20, 40 and 80 pixels and the threshold among $1/16$, $1/8$, $1/4$, $1/2$ and 1 event per pixel. The criterion is the bits that REMAKE saves against the subset schemes at equal fidelity for $K = 4$, as a share of the bits of all events. It selects $B = 20$ and $\psi = 1$ on both datasets. On DSEC the runner-up, $B = 40$ and $\psi = 1$, saves 0.086 of the bits of all events against 0.090. The configuration is then fixed, and every result below is over the other 100 eTraM recordings and 36 DSEC sequences. The selected threshold is the largest candidate. The saving grows with the threshold over the whole range while the fidelity falls, so the threshold acts as a rate control, and Section VI-D reports the whole range. Only B and ψ are calibrated, and T_0, K, the grids and the coders are set in advance. The regions, the key events and the maps are computed from the stream, for every group anew. With the transport coder all events cost 23.4, 24.0 and 22.0 bits each by day, by night and on DSEC, and with the matched coder 1.56, 2.04 and 1.64. A receiver that counts on the working grid thus needs 7% to 9% of the bits of the transport, and we take the matched coder as the reference.

Statistics. A recording enters with the sums over its groups, and we report the median over recordings, with a 95% interval from a bootstrap over recordings where one is given. Of the 100 eTraM recordings, 93 hold the ten groups that we require (53 by day and 40 by night, 4958 groups). The DSEC sequences are shorter, so we require five groups there, and all 36 enter, with 1189 groups.

B. The Three Regimes

Fig. 2 compares Theorem 2 with a rate-distortion function that is computed without it, for one pixel with $n = 3$ and $\sigma/\tau_i = 0.05$. These values illustrate the theorem and are not those of a particular sensor. By (35), the event times are an orthogonal image of the scalar X_i, which carries the phase, and of $n - 1$ normal variables. The iteration of [26] gives the rate-distortion function of X_i on a grid of 1217 points, and the normal components enter in closed form at the same slope.

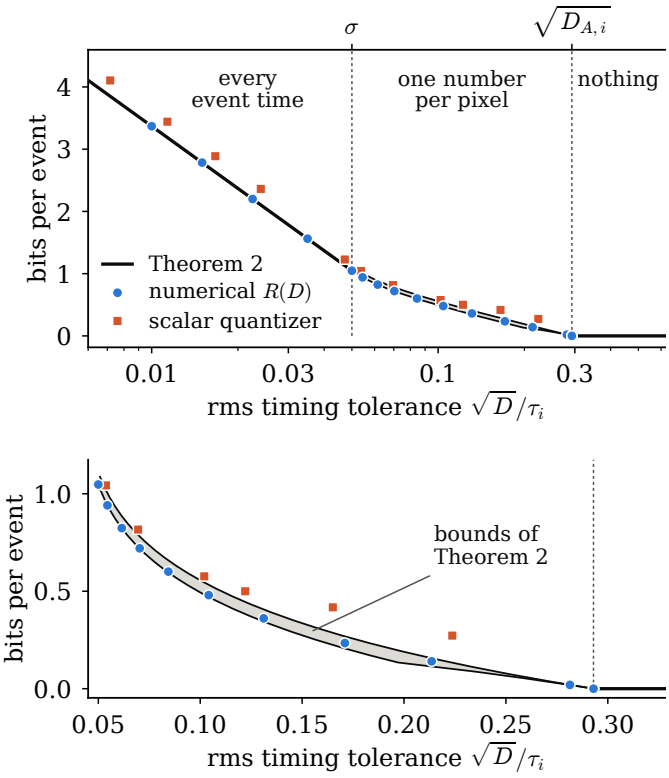


Fig. 2. Rate of one pixel against the timing tolerance for $n = 3$ and $\sigma/\tau_i = 0.05$: Theorem 2, a numerical rate-distortion function and a scalar quantizer. Bottom: the regime between σ and $\sqrt{D_{A,i}}$.

Below σ^2 the numerical rate equals (17) to within 4×10^{-6} nat. Between σ^2 and $D_{A,i}$ it stays inside the band between the bounds (18) and (21), which is at most $\Gamma_i = 0.187$ bit per pixel wide here, against the limit of 0.255 bit. The rate leaves σ^2 on the lower bound $U_i - \Gamma_i$: the phase is still described finely there, and the code profits fully from its distribution, which is uniform and not normal. Toward $D_{A,i}$ the description becomes coarse, the rate depends on the phase only through its variance, and the numerical rate, the bound (21) and U_i meet. From $D_{A,i}$ on the rate is zero, and the regenerated events (22) of 2×10^6 simulated pixels have the distortion $0.9994\, D_{A,i}$.

The squares are a uniform scalar quantizer with entropy coding, applied to what each regime describes. On every component of the event times it stays 0.254 to 0.258 bit per event above (17). This is the loss $\frac{1}{2}\ln(2\pi e/12)$ of a uniform quantizer at fine resolution, the difference between the entropy $h - \ln\Delta$ of the index of a quantizer of step Δ and (17) at the distortion $\Delta^2/12$. On the mean event time alone, with the other components left out, it stays 0.09 to 0.19 bit per event above the lower bound up to $D_{A,i}$. A code that describes every event time, then one number per pixel, then nothing, is therefore within 0.26 bit per event of the limit in all three regimes.

C. From Regeneration to REMAKE

Between the decoder (22) and REMAKE lie four steps: the events are moved instead of regenerated, the map moves them by whole pixels, a window boundary cuts the event trains, and the map is fitted to the events. Fig. 3 separates them

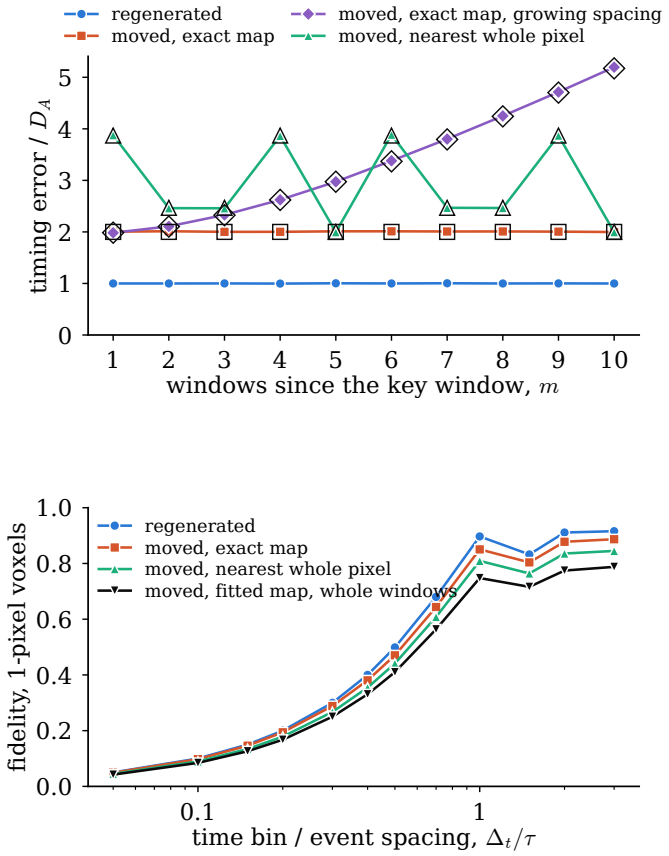


Fig. 3. From the regenerating decoder to REMAKE on the model, $n = 3$, $\sigma/\tau = 0.05$, 10 or 10.4 pixels per window, spacing growing by 5% per window where marked. Top: timing error per event against the window m; open markers give Proposition 2 with the bias of a whole-pixel map. Bottom: fidelity on one-pixel voxels against the bin; dotted lines give (33) and (34).

on the model, for an edge that passes ten or 10.4 pixels per window, so that the exact map is or is not a whole number of pixels. In timing error, the regenerated events reach D_A and the events moved by the exact map $2.006\,D_A$, against the $2D_A$ of Proposition 2. When the event spacing grows by 5% per window, the error of the moved events grows with m and averages $3.56\,D_A$, as the last term of (29) predicts. A map in whole pixels adds the squared timing bias of its rounding, which depends on the fractional part of the displacement and averages $0.92\,D_A$ at 10.4 pixels per window (Fig. 3, top). In voxel fidelity at $\Delta_t = \tau/2$, the regenerated events reach 0.499 and the moved events 0.470, against 0.5 and 0.472 from (33) and (34). The map in whole pixels lowers the fidelity to 0.441. Scored on whole windows, the exact map reaches 0.430, because a window boundary cuts the train of a source pixel and the train of its target at different events, and the fitted map reaches 0.410 at 10.4 pixels per window (Fig. 3, bottom). The fit picks the nearest whole pixel for 93% of the shifts at ten pixels per window and 78% at 10.4, because its working grid of two pixels and $T_0/3$ rarely separates neighboring shifts. The steps that REMAKE adds to moving, maps in whole pixels and windows that cut trains, thus lower the fidelity more than the phases of the moved events do.

TABLE I

REMAKE ON THE WORKING GRID, MATCHED CODER. b AND b_{sub} IN BITS PER EVENT, G_{rep} FOR THE REPEATED KEY EVENTS, WITH THE RATIO TAKEN PER RECORDING, G_{T} WITH THE TRANSPORT CODER.

	K	F	b	b_{sub}	G	G/G_{rep}	G_{T}
eTraM, day	4	0.71	0.42	0.65	1.55	2.57	2.68
	11	0.62	0.16	0.53	3.26	4.13	5.96
eTraM, night	4	0.65	0.35	0.57	1.62	1.82	2.85
	11	0.57	0.13	0.48	3.64	2.45	6.62
DSEC	4	0.67	0.52	0.66	1.31	1.48	2.38
	11	0.58	0.20	0.55	2.71	1.59	5.20

D. Bits at Equal Fidelity

Table I and Fig. 4 (left) give the bits of REMAKE against those of the subset schemes at its fidelity.

Gain. On eTraM a group of four windows costs 0.42 bit per event by day and 0.35 by night and reaches the fidelities 0.71 and 0.65. The cheapest subset scheme needs 0.65 and 0.57 bit at these fidelities, so $G = 1.55$ (95% interval 1.52 to 1.57) by day and 1.62 (1.61 to 1.65) by night. On DSEC $G = 1.31$ (1.29 to 1.34), and 1.32 over the 31 sequences from Zurich. The key events that REMAKE sends are 21% of the events of the group by day, 17% by night and 22% on DSEC, and the receiver rebuilds the rest. The gain grows with the group, to 3.26, 3.64 and 2.71 for eleven windows, at fidelities lower by 0.09, 0.08 and 0.10, with 7%, 6% and 8% of the events sent. A further window costs REMAKE only its maps, 3% of a record for $K = 4$ and 8% to 10% for $K = 11$, and costs a subset scheme every event that it keeps. For $K = 2$ REMAKE loses, with $G = 0.92$, 0.94 and 0.81. Half of such a group is its key window, and a key event is coded at one pixel, because a map moves it by whole pixels, where a subset is coded at the two pixels of the grid. From four windows on, the bits that the maps save exceed this cost.

Threshold. The threshold sets the operating point between bits and fidelity (Fig. 4, right). From $\psi = 1/16$ to $\psi = 1$, the bits for $K = 4$ fall from 0.62 to 0.42 by day, from 0.74 to 0.35 by night and from 0.72 to 0.52 on DSEC, the fidelity falls from 0.75 to 0.71, from 0.74 to 0.65 and from 0.71 to 0.67, and G rises from 1.17 to 1.55, from 0.93 to 1.62 and from 1.06 to 1.31. The blocks that a lower threshold adds cost key events and return few pairs in the later windows. Along the threshold REMAKE stays below the subset schemes in bits, except at night for $K = 4$ and $\psi \le 1/8$, where the gain is 0.93 and 1.01. At $\psi = 1$ the regions are 1% of the blocks of eTraM and hold 83% of the events of the key window by day and 69% by night. On DSEC they are 31% of the blocks and hold 86%.

Source of the gain. Key events that are repeated in place, without maps, stay below the subset schemes for $K = 4$, with $G_{\mathrm{rep}} = 0.60$, 0.90 and 0.88. The gain of REMAKE is therefore the gain of its maps, a factor 2.57, 1.82 and 1.48 over repetition (the median of the ratio per recording). Repetition does better where true events stay in the regions. Ten windows after the key window, 25% of the true events of eTraM by day still fall into the regions, and 70% on DSEC, where the whole scene moves. A repeated event there meets some other event

Figure file `figs/fig_bits_pair.pdf`
(directive 015)

Fig. 4. Bits as a percentage of the bits of all events, matched coder, working grid. Left: REMAKE and the cheapest subset scheme at the fidelity F of REMAKE, at $\psi = 1$, for groups of K windows; the labels give G. Right: the same against F for the five thresholds, $K = 4$ and 11.

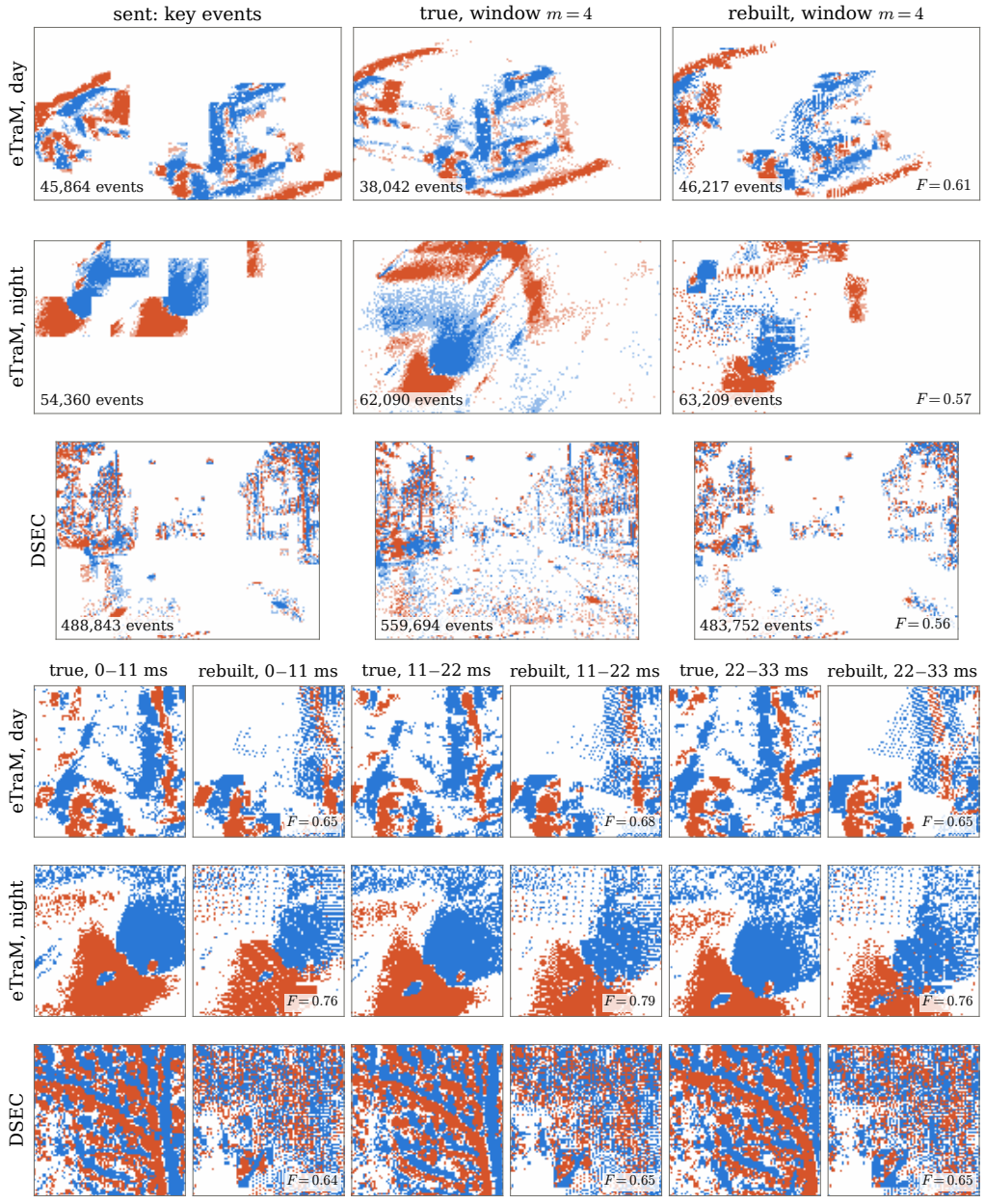


Fig. 5. A typical group of each stratum, signed event counts per pixel, positive minus negative polarity. Top: the key events that REMAKE sends, the true events of window $m = 4$, and the events that the receiver rebuilds for it from the stream. Bottom: an 80×80-pixel square of window $m = 4$ in its three time bins, true and rebuilt, with the fidelity of each bin.

in its voxel, and the fidelity of repeated windows falls only from 0.39 to 0.26 over ten windows on DSEC, against 0.18 to 0.07 on eTraM by day. Such pairs do not need the right motion, and the ratio G/G_{rep} removes them.

Rebuilt events. Fig. 5 shows one group of each stratum. Its recording has the median fidelity of window $m = 4$ in the stratum, with the fidelity of a recording formed from the sums over its groups, and the group is the one whose fidelity is closest to that of its recording: 0.61 by day, 0.57 by night and 0.56 on DSEC. Where a region holds key events, the rebuilt

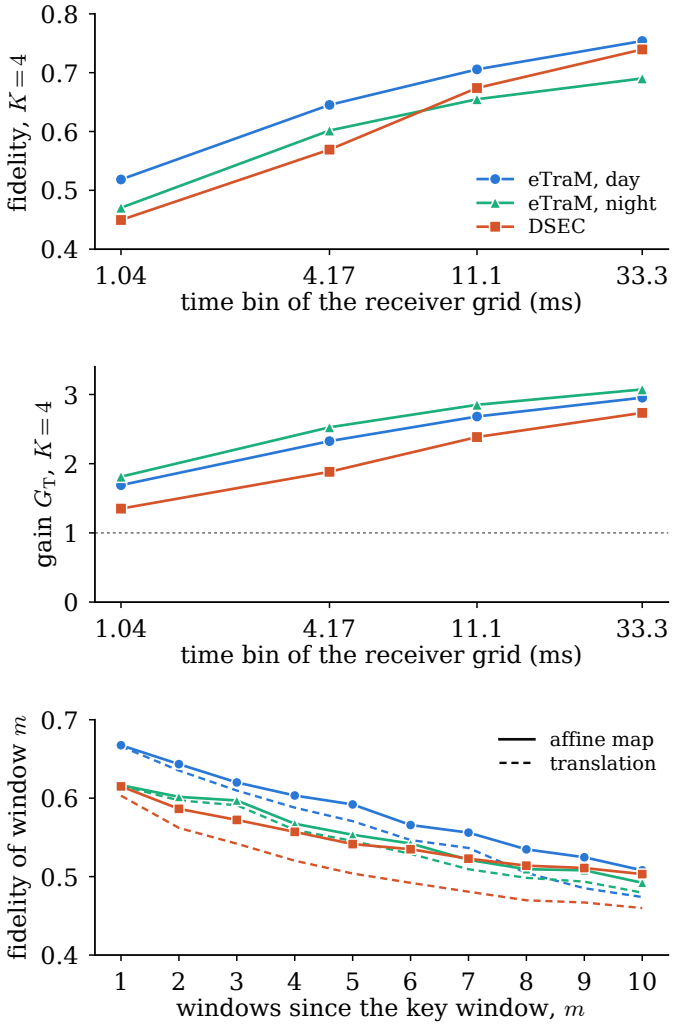


Fig. 6. Top: fidelity of a group of four windows against the time bin of the receiver grid, at $\Delta_x = 2$ pixels. Middle: the gain on the same grids, transport coder. Bottom: fidelity of window m on the working grid, for the affine map (solid) and the translation (dashed).

edges lie where the true ones are. Where none does, nothing is rebuilt, as for the road in the lower half of the DSEC view, whose blocks hold fewer than ψB^2 events in the key window. Inside the regions the rebuilt edges are dotted where the true ones are filled, since a map moves every key event to one pixel, and a map that enlarges the region spreads the same events over more pixels. The square of 80×80 pixels at the bottom of Fig. 5 is the one with the most true events of the window, and it holds about as many rebuilt as true events, so regions cover it. Its fidelity in each third of the window is 0.65 to 0.68 by day, 0.76 to 0.79 by night and 0.64 to 0.65 on DSEC, above the fidelity of the whole window, and it does not fall from one third to the next: inside the regions the rebuilt events follow the true ones in time at the resolution of the working grid.

E. Limits of the Gain

Time bin. The time bin of the receiver grid plays the role of the tolerance. From a bin of 33.3 ms to one of 1.04 ms, the fidelity of a group of four windows falls from 0.75 to 0.52 on eTraM by day, from 0.69 to 0.47 by night and from 0.74 to 0.45 on DSEC, and the gain with the transport coder falls from 2.95, 3.07 and 2.73 to 1.69, 1.81 and 1.35 (Fig. 6, top and middle). On the grid of one pixel and 1.04 ms it is 1.24, 1.42 and 1.01. By (33) and (34), neither regenerated nor moved events fill bins shorter than the event spacing beyond the ratio of the two, while a subset keeps its fidelity on every grid. This is the scaling of the threshold of Theorem 2. The event spacings of the recorded pixels were not measured, so the comparison is one of trends. The same trend separates objects. With annotated vehicles as regions and one translation per vehicle, the fidelity for $K = 4$ on the working grid rises from 0.52 below 20 pixels per second to 0.72 above 100 by day, and from 0.61 to between 0.71 and 0.74 by night: by (8) a faster edge has a shorter event spacing, so the same bin is a larger tolerance for it.

TABLE II
THE AFFINE MAP AGAINST ONE TRANSLATION PER BLOCK, MATCHED CODER, WORKING GRID. MAP: THE SHARE OF THE BYTES OF A RECORD THAT THE AFFINE MAPS TAKE FOR $K = 11$.

	G, $K = 4$		G, $K = 11$		F, $K = 11$		map
	affine	transl.	affine	transl.	affine	transl.	
eTraM, day	1.55	1.57	3.26	3.32	0.62	0.60	10%
eTraM, night	1.62	1.65	3.64	3.75	0.57	0.57	8%
DSEC	1.31	1.27	2.71	2.63	0.58	0.54	8%

Distance from the key window. The fidelity of window m falls with m, from 0.67 to 0.51 on eTraM by day, from 0.62 to 0.49 by night and from 0.62 to 0.50 on DSEC over ten windows (Fig. 6, bottom). The number of rebuilt events does not cause the fall: its ratio to the number of true events moves from 0.83 to 0.80 by day and from 0.86 to 0.83 on DSEC. The moved events fall into the wrong voxels. Two losses are present from the first window on, where the fidelity is 0.62 to 0.67: the events outside the regions, which hold the key window itself to 0.91, 0.82 and 0.92, and the timing of the moved events, which Section VI-C decomposes.

Motion model. The affine map raises the fidelity more than the gain (Table II). With one translation per block, the fidelity of window m is lower by up to 0.03 by day, 0.01 by night and 0.04 on DSEC at the tenth window, as the residual of a block translation against the flow accumulates. On eTraM the translation still gains 1% to 3% more, because the shape parameters cost more bits than their pairs return. On DSEC the whole scene moves with the camera: ten windows after the key window the median block has moved by 38 pixels, 91% of the blocks have a nonzero shape parameter, and the affine map gains 2.5% more for $K = 4$. The affine flow of the theory is thus the right model for a moving camera, and a field of block translations approximates it well enough for a static one.

F. Lossless Cost

By Theorem 1, a lossless code must describe the phase and the jitter of every pixel even when the motion is known. A recording does not give that conditional entropy, but it shows

how much motion saves for a given coder. For the events inside annotated vehicles of eTraM that are isolated and move at 20 to 300 pixels per second, with addresses of one pixel and times of one microsecond, the ideal code length under a density that is uniform over the box of the vehicle and the window is 18.3 bits per event. A density built from the events of the preceding 100 ms saves 0.5 bit by day and 1.2 by night when these events are left in place, and 2.9 and 3.0 bits when they are moved along the estimated motion. Motion thus saves about a sixth of the lossless cost for these densities. The remaining 15 bits bound the conditional cost of the events from above. They contain the private part of Theorem 1 and what these densities fail to model, and the measurement does not separate the two.

VII. Conclusion

The pixels on the path of a moving edge share the edge and its motion, and each has a threshold phase and a jitter of its own, which a lossless code must describe at every pixel. A timing tolerance removes first the jitter and then the phase from what a code must describe, and beyond a threshold set by the event spacing the events are regenerated from the edge and the motion alone. REMAKE carries this to a sender that has only the events: it moves key events along affine maps on a block grid. Moving instead of regenerating doubles the timing error but lowers the fidelity of a voxel receiver only by a term of order $1/n$. Calibrated on a few recordings of each dataset, REMAKE needs fewer bits than schemes that send a subset of the events at equal fidelity once a group holds four windows, on a static camera and on a moving one, and its advantage shrinks as the time bin of the receiver shortens, in line with the threshold of the theory. Two steps remain open. REMAKE moves a fixed number of events, keeps its regions for a whole group, and keeps its group length and threshold for a whole stream, while a change of scale changes the count, new objects enter, and the best group length and threshold vary with the scene. The theory covers one ramp edge in uniform passage, so texture, noise events and threshold mismatch remain to be modeled.

Appendix A
Proofs of Lemmas 1 and 2

Proof of Lemma 1. The reference starts in $a_i + C\mathbb{Z}$ by the definition of the phase. By (A1) it changes by $\pm C$, so it stays in this set, and an event requires $\ell_i = \mu_i \pm C$. □

Proof of Lemma 2. By (A2) and (A2′), ℓ_i is nondecreasing, equal to 0 before t_i, to $\gamma r_i(t - t_i)$ during the passage, and to V after it. Since $\mu_i \le 0 = \ell_i$ before the edge arrives and $\mu_i = a_i - C$, the events occur when ℓ_i reaches the levels $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as the level is at most $V = nC$. Because $0 < a_i \le C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma r_i(t - t_i) = a_i + (k-1)C$ gives (10), and a_i uniform on $(0, C]$ makes A_i uniform on $(0, \tau_i]$. □

Appendix B
Proof of Theorem 1

By Lemma 2, $T_i = c_i + A_i\mathbf{1} + \varepsilon_i$ with c_i a function of the template and the flow, $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and (A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n}\, A_i + \varepsilon_i^{\parallel}, \qquad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \tag{35}$$

where $\varepsilon_i^{\parallel} \sim \mathcal{N}(0, \sigma^2)$. The variables X_i and W_i are independent, because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i. Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (11).

Let f be the density of T_i and f_δ its average over the cells of the clock lattice. A cell has probability $\delta^n f_\delta$, so $H(Y_i \mid g, \phi) = n \ln(1/\delta) - \int f \ln f_\delta$ and $e_\delta = \int f \ln(f/f_\delta)$ is a relative entropy, which is nonnegative. The density f is a mixture over A_i of $\mathcal{N}(m, \sigma^2 I_n)$ densities and f_δ is the same mixture of their cell averages. By the joint convexity of relative entropy, e_δ is at most its largest value for a single $\mathcal{N}(m, \sigma^2 I_n)$ density, which is n times the value for a scalar normal density and depends only on δ/σ and on the offset of m in a cell. It tends to zero with δ/σ, uniformly in the offset [24, Ch. 8].■

Appendix C
Proof of Theorem 2

The distortion (14) is invariant under the translation by c_i and the orthogonal map Q, so R_i is the rate-distortion function of the independent components (X_i, W_i) of (35) under the total squared error nD. Their variances are s_i^2 and σ^2, the latter $n-1$ times. We drop the index i and write η for $\eta(\sqrt{n}\,\tau_i, \sigma)$ and $\ln^+ x = [\ln x]^+$.

Lower bound (17) *and* (18). Let $(\hat{X}, \hat{W})$ have component errors d_X and $d_1, \dots, d_{n-1}$ with $d_X + \sum_j d_j \le nD$. Independence of the components gives $I(X, W; \hat{X}, \hat{W}) \ge I(X; \hat{X}) + \sum_j I(W_j; \hat{W}_j)$. For a Gaussian component, $I(W_j; \hat{W}_j) \ge \frac{1}{2} \ln^+(\sigma^2/d_j)$, and $I(X; \hat{X}) \ge h(X) - h(X - \hat{X}) \ge \eta - \frac{1}{2} \ln(2\pi e\, d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d, so the Gaussian terms are smallest when all d_j equal a common d_W, and $d_W > \sigma^2$ brings no benefit. For $n \ge 2$ this gives

$$R(D) \;\ge\; \min_{d_W} \Big[\eta - \tfrac{1}{2} \ln\big(2\pi e\,(nD - (n-1)d_W)\big) + \tfrac{n-1}{2} \ln \frac{\sigma^2}{d_W} \Big], \tag{36}$$

with the minimum over $0 < d_W \le \sigma^2$, $d_W < nD/(n-1)$. The bracket is convex in d_W and its derivative vanishes only at $d_W = D$. The minimum is at $d_W = D$ when $D \le \sigma^2$, where it equals the right side of (17), and at $d_W = \sigma^2$ otherwise, where it equals $U(D) - \Gamma$. For $n = 1$ there is no W and the same values follow with $d_X = D$.

Equality for $D \le \sigma^2$. Write $\varepsilon = \varepsilon' + Z$ with independent $\varepsilon' \sim \mathcal{N}(0, (\sigma^2 - D) I_n)$ and $Z \sim \mathcal{N}(0, D I_n)$, and let $\hat{T} = c + A\mathbf{1} + \varepsilon'$. Then $T = \hat{T} + Z$ with Z independent of $\hat{T}$, the distortion is D, and $I(T; \hat{T}) = h(T) - h(Z)$, which is the lower bound.

Upper bound for $\sigma^2 < D < D_A$. Spend no rate on W, at error σ^2 per component, and describe X at error $d = nD - (n-1)\sigma^2 \in (\sigma^2, s^2)$ through $\hat{X} = (1 - d/s^2)(X - \mathbb{E}X + Z') + \mathbb{E}X$, with Z' an independent normal variable of variance $s^2 d/(s^2 - d)$. The error is d, and $I(X;\hat{X}) = I(X; X + Z') \le \frac{1}{2}\ln(s^2/d)$, because a normal input maximizes the mutual information across an additive normal noise channel at a given input variance [24].

Lower bound (21) *and zero rate.* For a variable Ξ with $\ln \mathbb{E}\, e^{\lambda(\Xi - \mathbb{E}\Xi)} \le \lambda^2\varsigma^2/2$ for all λ under a law P, and any law Q, the nonnegativity of the relative entropy between Q and the law with density proportional to $e^{\lambda\Xi}$ with respect to P gives $D(Q\|P) \ge \lambda(\mathbb{E}_Q\Xi - \mathbb{E}_P\Xi) - \lambda^2\varsigma^2/2$, and the best λ gives $(\mathbb{E}_Q\Xi - \mathbb{E}_P\Xi)^2 \le 2\varsigma^2 D(Q\|P)$. With P the law of X and Q its conditional law given $\hat{T}$, averaging over $\hat{T}$ yields $\operatorname{Var}\mathbb{E}[X \mid \hat{T}] \le 2\varsigma^2 I(X;\hat{T})$. The uniform part of X has width $a = \sqrt{n}\,\tau$ and satisfies the condition with $a^2/12$, since $\ln(\sinh x/x) \le x^2/6$, so X satisfies it with $\varsigma^2 = s^2$, and each W_j with $\sigma^2 \le s^2$. Since the conditional mean minimizes the squared error and the components are independent,

$$\begin{aligned} nD &\ge \mathbb{E}\|T - \mathbb{E}[T \mid \hat{T}]\|^2 \\ &= nD_A - \operatorname{Var}\mathbb{E}[X \mid \hat{T}] - \sum_j \operatorname{Var}\mathbb{E}[W_j \mid \hat{T}] \\ &\ge nD_A - 2s^2 I(T;\hat{T}), \end{aligned} \tag{37}$$

which is (21). With no description the best output is the mean, which is (22), at distortion $\frac{1}{n}(s^2 + (n-1)\sigma^2) = D_A$, so (19) holds. ∎

Appendix D
Proof of Theorem 3

The vectors T_i are independent and the distortion is additive over i. For any $\hat{T}$ with average distortion at most D, let $D_i = \mathbb{E}\, d(T_i, \hat{T}_i)$. Then $I\big((T_i)_i; (\hat{T}_i)_i\big) \ge \sum_i I(T_i;\hat{T}_i) \ge \sum_i R_i(D_i)$ with $\frac{1}{M}\sum_i D_i \le D$. Conversely, independent test channels whose rates are within any given margin of $R_i(D_i)$ come within M times that margin of the sum. Each R_i is convex, nonincreasing and continuous, and tends to infinity as $D_i \to 0$ by (17), so the minimum in (23) exists.

For $D < \sigma^2$, every R_i is differentiable at D with the derivative $-n/(2D)$ by (17), and it is convex on $(0,\infty)$. Hence $R_i(D_i) \ge R_i(D) - \frac{n}{2D}(D_i - D)$ for every $D_i > 0$, and summing over i with $\sum_i (D_i - D) \le 0$ shows that the allocation $D_i = D$ is optimal. This gives (24) for $D < \sigma^2$, and the case $D = \sigma^2$ follows by continuity.

By Theorem 2, $\sum_i R_i(D_i) = 0$ if and only if $D_i \ge D_{A,i}$ for every i, and such an allocation with average at most D exists if and only if $D \ge \frac{1}{M}\sum_i D_{A,i} = \bar{D}_A$. Finally, (21) and $s_i^2 \le n\tau_{\max}^2/12 + \sigma^2$ give $\sum_i R_i(D_i) \ge n\sum_i [D_{A,i} - D_i]^+/(2(n\tau_{\max}^2/12 + \sigma^2))$, and $\sum_i [D_{A,i} - D_i]^+ \ge M(\bar{D}_A - D)$, which is (26). ∎

Appendix E
Proof of Proposition 1

Divide side j of the box into the smallest number of equal cells of length at most $\varsigma_j = \sqrt{\xi}\,\sigma/(q\,S_j M^{\beta_j})$, and let $\hat{\theta}$ be the center of the cell that contains θ. Side j has at most $L_j M^{\alpha_j}/\varsigma_j + 1$ cells, so the index of $\hat{\theta}$ takes $\sum_j \ln\big(L_j S_j q\, M^{\alpha_j+\beta_j}/(\sqrt{\xi}\,\sigma) + 1\big)$ nats, which is (27). The decoder outputs $\hat{T}_{i,k} = \bar{t}_{i,k}(\hat{\theta})$. The segment from θ to $\hat{\theta}$ lies in the box, so by the mean value theorem $e_{i,k} = \bar{t}_{i,k}(\theta) - \bar{t}_{i,k}(\hat{\theta})$ satisfies $|e_{i,k}| \le \sum_j S_j M^{\beta_j}\varsigma_j/2 \le \sqrt{\xi}\,\sigma/2$. By (10), $T_{i,k} - \bar{t}_{i,k}(\theta) = A_i - \tau_i/2 + \varepsilon_{i,k}$ has mean zero given θ and variance $D_{A,i}$, and $e_{i,k}$ is a function of θ. Hence $\mathbb{E}\,(T_{i,k} - \hat{T}_{i,k})^2 = D_{A,i} + \mathbb{E}\, e_{i,k}^2 \le D_{A,i} + \xi\sigma^2/4$, and the average over the path is at most $\bar{D}_A + \xi\sigma^2/4 \le (1+\xi)\bar{D}_A$. ∎

Appendix F
Proof of Proposition 2

By (10) and (A4), $T_{i,k} - \tilde{T}_{i,k} = (A_i - A_j) + (k-1)(\tau_i - \tau_j) + \varepsilon_{i,k} - \varepsilon_{j,k}$. The phases and the jitter of the two pixels are independent, so this difference has the mean $(k - \frac{1}{2})(\tau_i - \tau_j)$ and the variance $\tau_i^2/12 + \tau_j^2/12 + 2\sigma^2 = D_{A,i} + D_{A,j}$. Its mean square is the variance plus the squared mean, and $\frac{1}{n}\sum_{k=1}^{n}(k - \frac{1}{2})^2 = (4n^2 - 1)/12$. ∎

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